



PDF Download
3769102.3774631.pdf
15 February 2026
Total Citations: 0
Total Downloads: 189

 Latest updates: <https://dl.acm.org/doi/10.1145/3769102.3774631>

RESEARCH-ARTICLE

Toward Design of a Scalable Federated Unlearning Framework for Trustworthy Edge Intelligence

HAITHAM Y ADARBAH, Texas A&M University-Kingsville, Kingsville, TX, United States

KEWEI SHA, University of North Texas, Denton, TX, United States

AFZEL NOORE, Texas A&M University-Kingsville, Kingsville, TX, United States

Open Access Support provided by:

Texas A&M University-Kingsville

University of North Texas

Published: 03 December 2025

[Citation in BibTeX format](#)

SEC '25: Tenth ACM/IEEE Symposium on
Edge Computing
December 3 - 6, 2025
VA, Arlington, USA

Conference Sponsors:
SIGMOBILE

Toward Design of a Scalable Federated Unlearning Framework for Trustworthy Edge Intelligence

Haitham Y. Adarbah*

Department of Electrical Engineering
and Computer Science, Texas A&M
University–Kingsville
Kingsville, TX, USA
Haitham.Adarbah@tamuk.edu

Kewei Sha

Department of Data Sciences,
University of North Texas
Denton, TX, USA
Kewei.Sha@unt.edu

Afzel Noore

Department of Electrical Engineering
and Computer Science, Texas A&M
University–Kingsville
Kingsville, TX, USA
Afzel.Noore@tamuk.edu

Abstract

Federated learning (FL) enables collaborative model training across edge devices without centralizing raw data, but existing frameworks remain ill-equipped to support data privacy regulations mandated by GDPR, HIPAA, and CCPA. Once user data has influenced training, its verifiable removal becomes prohibitively expensive, particularly in non-IID and resource-constrained edge environments. This paper introduces a modular and scalable federated unlearning framework that unifies three complementary strategies: gradient subtraction, knowledge distillation, and checkpoint rollback, within an adaptive decision layer. A resource-aware checkpoint manager reduces storage costs through compression and pruning, while a privacy and trust layer integrates zero-knowledge proofs, differential privacy, and Merkle-based audit logs to provide verifiable guarantees of deletion. A non-IID-aware aggregator further preserves fairness across heterogeneous clients. Unlike prior approaches, our proposed framework systematically integrates rollback efficiency with formal privacy protections and auditability, offering a practical path toward trustworthy and regulation-compliant unlearning in domains such as healthcare, transportation, and smart agriculture.

Keywords

Federated Unlearning, Edge Intelligence, Differential Privacy, Zero-Knowledge Proofs (ZKPs), Trustworthy Artificial Intelligence

1 Introduction

As edge computing increasingly adopts federated learning (FL) [1] to enable collaborative model training across edge devices, concerns about data privacy, user autonomy, and regulatory compliance have intensified. Traditional FL architectures are ill-equipped to address scenarios where users request data deletion or revocation after training, leading to significant risks under regulations such as GDPR Article 17, CCPA 1798.105, and HIPAA (Health Insurance Portability and Accountability Act). In particular, once user data has influenced a model, its complete and verifiable removal remains computationally prohibitive, especially in resource-constrained, non-IID federated settings.

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. SEC '25, Arlington, VA, USA

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2238-7/25/12

<https://doi.org/10.1145/3769102.3774631>

Recent advances such as FedEraser [8] and SIFU [2] have attempted to address this gap via gradient rollback and efficient checkpointing, respectively, but both approaches face limitations in scalability, storage efficiency, and adaptability to real-world deployments. Meanwhile, VERiFi [3] and Fast-FedUL [5] offer promising directions for verification and lightweight edge-compatible unlearning but lack integration with robust architectural support and formal privacy guarantees, which are mathematically defined protections such as differential privacy and zero-knowledge proofs that enable provable, auditable data privacy and unlearning.

This work introduces a scalable federated unlearning framework that integrates three complementary removal strategies: gradient subtraction, knowledge distillation, and checkpoint rollback, within a modular system designed for edge AI environments. Our approach enhances trust and auditability through the use of zero-knowledge proofs (ZKPs) [15], differential privacy (DP) [1], and cryptographic audit logging, and is specifically built to support non-IID data and asynchronous client participation. The proposed framework directly addresses the technical and practical limitations of previous solutions and offers a deployable path forward for user-verifiable real-time unlearning in next-generation AI-driven wireless networks.

The rest of the paper is organized as follows. Section II presents the motivation. Section III reviews related work. Section IV details the proposed comprehensive framework. Section V discusses important open research challenges. Finally, Section VI concludes the paper and outlines future research directions.

2 Motivation

The adoption of FL in wireless edge networks has enabled collaborative training across heterogeneous and privacy-sensitive devices without centralizing raw data. However, as FL matures, new challenges arise concerning *user autonomy*, *data revocation*, and *regulatory compliance*. Once client contributions are integrated into global models, removing their influence becomes *computationally prohibitive* in practice, especially under non-IID data distributions and edge resource constraints [8].

This limitation creates a critical gap between FL deployments and real-world requirements. For instance, under GDPR Article 17 (“Right to be Forgotten”), users are legally entitled to request the removal of their data from learned models. Similarly, CCPA Section 1798.105 enforces consumer rights to deletion of personal information. In healthcare and transportation domains, where wireless edge

intelligence is rapidly deployed, failure to comply with such mandates can lead to severe privacy violations and regulatory penalties [4, 8].

Recent advances have taken initial steps toward *federated unlearning (FU)*. Approaches such as FedEraser (gradient rollback) [8], SIFU (sequential client unlearning) [2], and Fast-FedUL (training-free strategies) [5] offer partial solutions.

At the same time, trust in AI systems increasingly depends not only on model accuracy but also on their ability to provide *verifiable privacy guarantees* [9]. Techniques like zero-knowledge proofs (ZKPs) [15] and differential privacy (DP) [1] have shown promise for auditing and limiting residual data leakage. However, these have yet to be systematically integrated into FU frameworks optimized for edge environments.

Therefore, there is a pressing need for a scalable, verifiable, and resource-aware FU framework that addresses both technical challenges (rollback efficiency, communication overhead, non-IID aggregation) and societal imperatives (user autonomy, regulatory compliance, and auditability). This motivates our work, which introduces a modular, real-time framework designed for AI-driven edge networks where trustworthy unlearning is essential.

In this work, we present a comprehensive federated unlearning framework tailored for AI-driven edge networks. The framework integrates multiple strategies and verification mechanisms to ensure scalability, verifiability, and adaptability under non-IID data, asynchronous updates, and limited device resources.

The design is organized into modular components:

- (1) Unlearning Engine: Adaptively selects among gradient subtraction, knowledge distillation, and checkpoint rollback.
- (2) Checkpoint Manager: Compresses and prunes snapshots for efficient storage and fast rollback.
- (3) Privacy and Trust Layer: Combines ZKPs, differential privacy, and Merkle-based auditability for verifiable deletion.

Key research challenges remain, including provable guarantees under non-IID and asynchrony, adaptive checkpointing to balance accuracy and efficiency, scalable proof generation, fairness after deletions, and robustness against adversarial deletion requests.

3 Related Work

Federated unlearning has emerged as a critical extension of federated learning, enabling the removal of user data contributions from trained models [10, 11]. Early approaches such as FedEraser [8] introduced gradient rollback combined with checkpointing for client-level removal. While effective in principle, FedEraser suffers from high storage costs and retraining overhead. Building on this direction, SIFU [2] proposed sequential informed unlearning, which provides provable guarantees of client removal in federated optimization. However, its reliance on sequential client processing limits scalability in large-scale deployments. Fast-FedUL [5] explored a training-free approach inspired by influence functions, offering very fast unlearning with resilience to data skew, but at the expense of accuracy and without strong privacy protections.

Other research has emphasized verifiability and trust. VERiFi [3] demonstrated how lightweight cryptographic proofs can provide accountability in unlearning, yet it does not integrate stronger formal

privacy guarantees such as differential privacy (DP) [1] or zero-knowledge proofs (ZKPs) [15]. Meanwhile, recent advances like FedSSU [6] proposed decentralized unlearning based on saliency maps to reduce central overhead and support flexible client-side deletion. Similarly, FUSED [16], introduced at CVPR 2025, leverages sparse adapters for reversible knowledge overwriting, enabling efficient and selective unlearning but requiring sensitive-layer identification and extensive evaluation in computer vision tasks.

As summarized in Table 1, existing approaches advance the state of federated unlearning but remain constrained by three enduring gaps: (1) scalability under edge resource limitations, (2) lack of unified integration of verifiable privacy (e.g., ZKPs, DP, auditability), and (3) limited robustness to non-IID data distributions. Our proposed framework directly addresses these gaps by integrating multiple rollback strategies with lightweight checkpointing, while embedding formal privacy guarantees and cryptographic auditability tailored for wireless edge networks.

4 System Design

The proposed federated unlearning framework is designed to operate under the stringent constraints of wireless edge environments, while ensuring verifiable privacy guarantees and robustness to heterogeneous, non-IID client data. The architecture integrates five core modules: (1) an adaptive Unlearning Engine, (2) a resource-aware Checkpoint Manager, (3) a Privacy Module with formal guarantees, (4) an Audit and Verification Layer, and (5) a Non-IID-Aware Model Aggregator. Figure 1 illustrates the high-level workflow. In this section, we present the technical details of each component, including mathematical formulations, protocol steps, and complexity considerations.

4.1 Overview of the Proposed Framework

To meet the need for trustworthy data revocation in federated learning, we propose a scalable unlearning framework that integrates rollback strategies, formal privacy guarantees, and support for non-IID data in edge environments. The framework is designed for real-world constraints such as limited bandwidth, energy, and memory in mobile, vehicular, and IoT devices.

As shown in Figure 1, the workflow combines edge clients, three unlearning strategies, and a verification pipeline. Key modules include an adaptive Unlearning Engine, a resource-aware Checkpoint Manager, a Privacy Module, an Audit Layer, and a non-IID-aware Aggregator, together enabling efficient and verifiable unlearning.

The Unlearning Engine dynamically selects among gradient subtraction, knowledge distillation, and checkpoint rollback to balance precision and resource cost. The Checkpoint Manager captures compressed snapshots with adaptive pruning, reducing storage while supporting fast rollback. The Privacy Module embeds zero-knowledge proofs and differential privacy to provide verifiable and private deletion, while the Audit Layer records all operations with cryptographic logs for compliance. Finally, the Aggregator recalibrates models after deletions to preserve fairness across heterogeneous clients.

This modular architecture achieves scalable, low-latency unlearning suited to applications in healthcare, transportation, and smart agriculture.

Table 1: Comparison of Key Federated Unlearning Approaches

Paper	Core Strategy	Strengths	Limitations
FedEraser (IWQoS 2021) [8]	Gradient rollback + checkpointing	Direct client-level removal; first practical FU framework	Requires storage of intermediate states; retraining overhead for large models
SIFU (AISTATS 2024) [2]	Sequential informed unlearning	Provable guarantees of client removal; efficient in sequential settings	Sequential processing limits scalability in large deployments
Fast-FedUL (ECML PKDD 2024) [5]	Training-free, influence-function inspired	Extremely fast; resilient to skewed data	Accuracy degradation in complex tasks; lacks strong privacy guarantees
VERiFi (IEEE TDSC 2024) [3]	Verifiable unlearning via cryptographic proofs	Builds trust and accountability; lightweight verification	No integration of DP or ZKPs; focuses on verification not rollback efficiency
FedSSU (J. Supercomputing 2025) [6]	Decentralized unlearning via saliency maps	Scalable; supports flexible client-side unlearning; reduces central overhead	Focuses on similarity-based filtering; less effective for rollback/gradient-based cases
FUSED (CVPR 2025) [16]	Sparse adapters (knowledge overwriting)	Enables reversible unlearning; efficient layer-specific adaptation	Requires sensitive-layer identification; mainly evaluated in vision tasks
Our Framework	Hybrid: gradient subtraction, distillation, checkpoint rollback + ZKPs + DP	Modular; resource-aware; non-IID support; verifiable privacy and auditability	Requires adaptive decision policies; efficiency depends on checkpoint compression

4.2 Unlearning Engine

At the heart of the system lies the Unlearning Engine, which executes one of three strategies: gradient subtraction, knowledge distillation, or checkpoint rollback. The choice is dynamically made by a decision logic that balances precision, latency, and resource availability.

Gradient Subtraction: Gradient rollback methods, as introduced in FedEraser [8], approximate unlearning by removing historical gradient contributions from client data. Suppose the global model at round t is parameterized by w_t . If client u requests deletion, its contribution can be approximated as:

$$w_{t+1} = w_t - \eta \nabla L(D_u) \quad (1)$$

where η is the learning rate. This operation has $O(d)$ complexity, where d is the number of parameters. While effective for small deletions, it becomes costly if gradient histories are unavailable.

Knowledge Distillation: Following approaches used in lightweight FU frameworks [5], our system employs a teacher–student strategy to transfer knowledge after deletions. The teacher is the global model prior to deletion, and the student is retrained without the deleted contribution. The distillation loss is:

$$L_{KD} = \alpha L_{CE}(y, \hat{y}) + (1 - \alpha) \tau^2 KL(\sigma(z/\tau) \| \sigma(\hat{z}/\tau)) \quad (2)$$

where z and $\hat{z}$ are logits of teacher and student models, τ is the temperature, and α balances cross-entropy and distillation. This avoids storing all historical gradients [14].

Checkpoint Rollback: Efficient checkpointing, explored in SIFU [2], provides a scalable way to reverse updates. In our system,

rollback restores w_{t-k} from compressed snapshots. The trade-off is rollback granularity versus storage overhead, motivating adaptive checkpointing.

Decision Logic: Strategy selection is formalized as:

$$\min_{s \in \{GS, KD, CR\}} C_s(R, A) \quad (3)$$

where C_s is the estimated cost of strategy s given device resources R and accuracy tolerance A . This aligns with adaptive FU decision layers proposed in recent verification frameworks such as VERiFi [3].

4.3 Checkpoint Manager

Relation to Unlearning Strategies: While the Checkpoint Manager is most directly tied to the checkpoint rollback strategy, its utility extends across all three unlearning methods. In rollback, compressed checkpoints enable fast restoration without retraining. For gradient subtraction, checkpoints act as a fallback when historical gradients are unavailable, providing a clean baseline for removing residual contributions. For knowledge distillation, checkpoints can serve as teacher models that guide the student model after deletion requests. Thus, the Checkpoint Manager is mandatory for rollback and complementary for subtraction and distillation, ensuring a unified and resource-aware support layer for all unlearning strategies.

To enable efficient rollback, the Checkpoint Manager maintains versioned model states with adaptive compression and pruning. Unlike naive storage of all rounds, our design triggers checkpoints

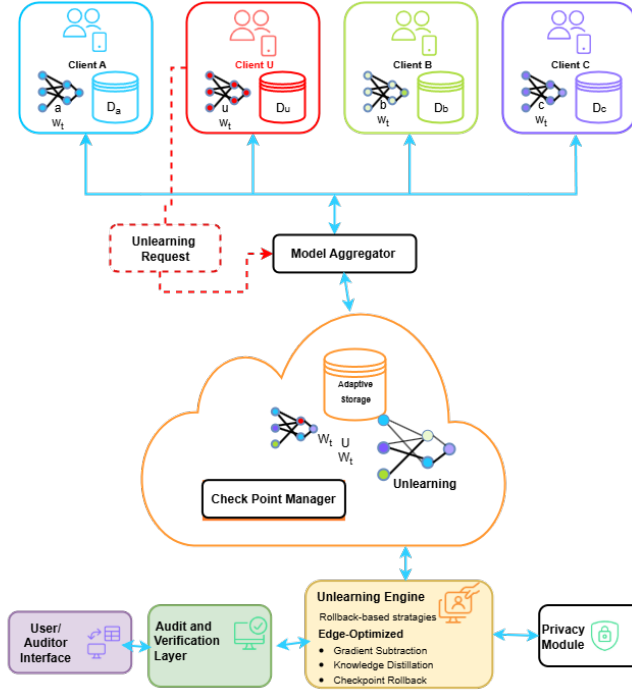


Figure 1: Scalable Federated Unlearning Framework Architecture

based on divergence thresholds:

$$\|w_t - w_{t-k}\|_2 \geq \delta \quad (4)$$

where δ is a divergence threshold. Compression techniques include quantization, low-rank factorization, and sketching. This reduces storage complexity from $O(Td)$ in naive approaches to $O(\log T \cdot d_c)$, where $d_c \ll d$ is the compressed dimensionality. Metadata tracking is decentralized, with a layered scheme separating client-specific versus global checkpoints, similar in spirit to FedSSU's decentralized tracking [6].

4.4 Privacy Module

Ensuring provable deletion requires integrating zero-knowledge proofs (ZKPs) and differential privacy (DP).

Verification: For every unlearning operation U , a proof π is generated such that:

$$\mathcal{V}(U, \pi) = 1 \quad (5)$$

if and only if U was correctly executed. This builds on verification methods such as VERiFi [3], but extends them with cryptographic guarantees from recent ZKP protocols [15].

Differential Privacy: DP is applied during rollback or aggregation to bound residual influence:

$$\Pr[M(D) \in S] \leq e^\epsilon \Pr[M(D') \in S] + \delta \quad (6)$$

where D and D' differ in one record. This builds on the formal framework of DP in FL [1], ensuring approximate unlearning even when rollback is partial.

4.5 Audit and Verification Layer

All unlearning events are logged in a tamper-resistant structure. We adopt Merkle trees, where each leaf encodes a deletion request and the root hash provides integrity. Given n operations, verification requires $O(\log n)$ time. Compared with blockchain approaches, our design avoids high energy costs while retaining verifiability, consistent with lightweight auditability discussed in VERiFi [3].

4.6 Non-IID-Aware Aggregator

After unlearning, the global model must be recalibrated to preserve fairness and accuracy across heterogeneous clients. We extend FedAvg with weighted aggregation:

$$w_{t+1} = \sum_i \frac{|D_i|}{\sum_j |D_j|} w_t^i \quad (7)$$

where weights are adjusted to account for distribution skew. Recent work such as FUSED [16] highlights the risks of imbalance when selective unlearning occurs in non-IID settings. To address this, we introduce a correction term:

$$w_{t+1} = w_{t+1} + \lambda \Delta_{fair} \quad (8)$$

where Δ_{fair} enforces balance across minority clients. Asynchronous updates are supported by buffering delayed gradients and applying staleness-aware scaling.

4.7 Workflow Integration

The complete workflow is summarized in Figure 1 and can be represented in Figure 2.

4.8 Evaluation Metrics

To validate the proposed framework, we define a set of metrics that capture both system-level efficiency and unlearning fidelity:

- **Unlearning Accuracy:** The distance between the unlearned model w_u and a model retrained from scratch without the deleted data w^* , following evaluation protocols in prior FU work [2, 16]:

$$\Delta_{acc} = \|w_u - w^*\|_2. \quad (9)$$

- **Latency:** Total request completion time:

$$T_{total} = T_{engine} + T_{checkpoint} + T_{verification} + T_{aggregation}. \quad (10)$$

- **Storage Overhead:** Extra memory cost relative to a baseline FL system:

$$S_{overhead} = \frac{S_{system} - S_{baseline}}{S_{baseline}}. \quad (11)$$

- **Communication Cost:** Bits transmitted per unlearning event, including rollback, proofs, and audit logs.
- **Energy Consumption:** Joules per unlearning request:

$$E = \alpha C_{cpu} + \beta C_{comm} \quad (12)$$

where C_{cpu} and C_{comm} represent computation and communication costs.

- **Verification Robustness:** Probability of forging or bypassing proofs, ideally negligible under ZKP soundness [15].

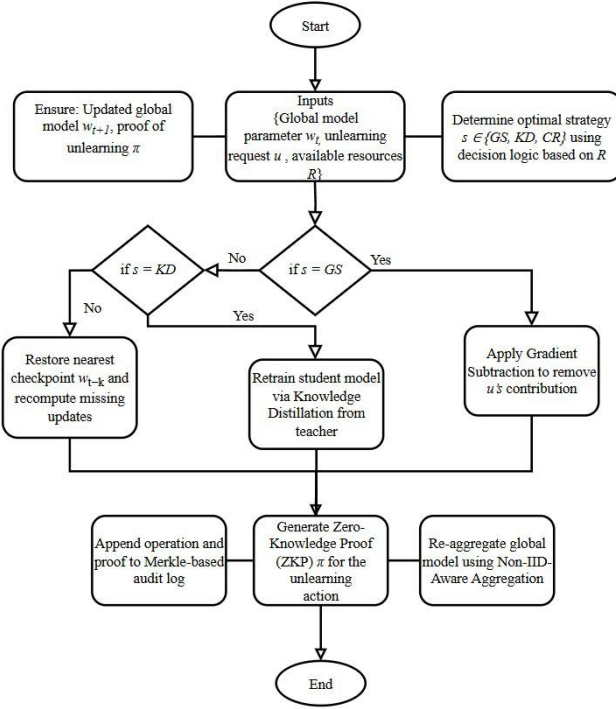


Figure 2: Federated Unlearning Workflow

These metrics will allow us to benchmark the framework against existing FU systems such as FedEraser [8], SIFU [2], Fast-FedUL [5], VERiFi [3], FedSSU [6], and FUSED [16]. By explicitly measuring accuracy, latency, storage, communication, energy, and verifiability, we provide a holistic assessment of both technical efficiency and trustworthiness.

5 Discussion and Open Problems

Our proposed framework demonstrates how federated unlearning can be made modular, verifiable, and resource-aware through the integration of multiple rollback strategies, adaptive checkpointing, and formal privacy guarantees. While this design provides a strong foundation, several open challenges remain that define a roadmap for future research.

5.1 Foundational Theory and Guarantees

Provable unlearning under non-IID and asynchrony: We need explicit bounds that relate each strategy’s output to a from-scratch retrain w^* , e.g., $\|w_u - w^*\|_2 \leq \epsilon$ with high probability, where ϵ depends on client heterogeneity and staleness. Such results would make Δ_{acc} predictable under realistic edge participation [2, 4]. The challenge is that current FU methods cannot guarantee that the unlearned model is mathematically close to a retrained-from-scratch baseline, especially with heterogeneous clients and delayed updates. Solving this would establish formal accuracy and stability guarantees for real deployments.

Fairness after deletions: Selective deletions can increase group disparities under non-IID data. Define a fairness drift metric (e.g., group loss gap) and use the aggregator’s correction term Δ_{fair} to keep disparity below a target while maintaining small Δ_{acc} . This is a constrained aggregation problem triggered after each unlearning event [16]. Selective unlearning can disproportionately harm minority or skewed client groups, amplifying bias. The challenge is to design fairness-aware aggregation mechanisms that re-balance models while preserving global accuracy.

Privacy-preserving knowledge distillation: Distillation avoids storing gradients but teacher logits can leak. Couple KD with DP noise and calibrate τ (temperature) and α (mixing) to achieve a given (ϵ, δ) at minimal accuracy loss, yielding a clear privacy-utility curve for KD-based unlearning on edge devices [14]. Right now, strategy choice is rule-based, but workloads are unpredictable. The challenge is learning adaptive policies that map resource states (CPU, bandwidth, energy) to strategies while respecting privacy and latency constraints.

5.2 Systems Design and Scalability

Adaptive, multi-objective checkpointing: The current divergence trigger δ should adapt online to minimize storage and energy while meeting accuracy/latency budgets. Concretely, choose δ_t and snapshot cadence to reduce $S_{overhead}$ and E subject to $\Delta_{acc} \leq \epsilon$ and $T_{total} \leq \tau$, using lightweight controllers or bandits driven by device/network telemetry [8]. The difficulty is balancing checkpoint frequency with limited memory and energy budgets: too frequent wastes resources, too sparse harms fidelity. A solution requires dynamic policies that tune checkpoint cadence in response to device and network conditions.

Hierarchical and cross-silo unlearning: Real deployments are multi-tier (device→edge→cloud). We must propagate deletions across tiers without redundant work, align checkpoints between tiers, and compose proofs end-to-end, targeting rollback latency that does not grow with the number of tiers [6]. Edge-to-cloud systems must propagate deletion requests across tiers, but redundant computation and proof verification increase latency. The challenge is to synchronize checkpoints and proofs across layers with minimal coordination cost.

Lightweight lineage and influence tracking: Targeted unlearning requires data-to-model lineage, but full histories are expensive. Maintain sketch-based influence summaries linked to the audit log so removals can be more precise than whole-client rollback while staying within storage and communication budgets [4]. Precise lineage is required to target which client contributions to remove, but storing full histories is infeasible. The challenge is to design compact, sketch-based influence tracking that enables selective unlearning under storage limits.

SLA-aware, real-time unlearning: Applications such as healthcare require service-level objectives (e.g., 95th-percentile latency). Design queueing and admission control (proof batching, urgency-aware prioritization) to meet T_{total} percentiles under load, with explicit accuracy and cost trade-offs [12]. Applications such as healthcare need strict latency service-level objectives (e.g., 95th percentile completion time). The challenge is to design queueing and scheduling

policies that prioritize urgent requests while balancing accuracy and energy use.

Hardware-aware implementations on edge accelerators: Co-design checkpoint formats (mixed-precision, low-rank/sparse) and rollback/distillation kernels for NPUs/GPUs on edge devices to reduce E and T_{engine} without increasing Δ_{acc} , validated on representative SoCs [13]. Most FU techniques are designed abstractly, ignoring device hardware. The challenge is to co-design checkpoint formats and rollback kernels to exploit NPUs/GPUs, lowering energy and latency while retaining fidelity.

5.3 Trust, Security, and Auditability

Scalable verification proof-carrying unlearning: ZK proofs certify correct unlearning, but proof time/size can dominate $T_{\text{verification}}$. We need recursive/incremental proofs that compose across rounds/clients and selective disclosure over Merkle logs, cutting verification cost by an order of magnitude without weakening soundness [15]. Zero-knowledge proofs are trustworthy but expensive; proof generation and verification can overwhelm latency targets. The challenge is to design recursive or incremental proofs that keep costs practical without weakening security.

Adversarial and abuse-resistant unlearning: Attackers may submit strategic deletion requests (e.g., removing rare-class contributors). We need influence-based monitoring and rate-limiting policies that protect utility and resource budgets while preserving legitimate rights, with actions recorded in the Audit/Verification layer [7]. Malicious users may repeatedly request deletions to degrade model quality or overload resources. The open challenge is building abuse-detection and throttling policies that preserve legitimate deletion rights but block harmful patterns

Addressing these challenges will require co-design across theory, systems, and trust layers, ultimately enabling federated unlearning to meet both technical and societal privacy requirements for trustworthy edge intelligence.

6 Conclusion

This paper presented a scalable federated unlearning framework designed specifically for edge networks. By unifying gradient subtraction, knowledge distillation, and checkpoint rollback under an adaptive decision layer, and by embedding zero-knowledge proofs, differential privacy, and cryptographic auditability, the framework achieves efficient, verifiable, and regulation-compliant unlearning. A resource-aware checkpoint manager minimizes storage overhead, while a non-IID-aware aggregator ensures fairness and efficiency across heterogeneous edge devices. Compared with existing approaches such as FedEraser, SIFU, Fast-FedUL, and VERiFi, our design advances the state-of-the-art by jointly addressing rollback efficiency, verifiability, and robustness to non-IID data. Looking ahead, we envision extending this foundation to energy-aware and hardware-optimized implementations, hierarchical unlearning, and scalable verification protocols suitable for real-world deployment. Our proposed framework takes a critical step toward standardizing federated unlearning in edge networks, bridging the gap between regulatory compliance and practical deployments. Future research will extend this foundation to next-generation healthcare, transportation, and smart agriculture applications.

References

- [1] A. El Oudrhiri and A. Abdelhadi. 2022. Differential privacy for deep and federated learning: A survey. *IEEE Access* 10 (2022), 22359–22380.
- [2] Y. Fraboni, M. Van Waerebeke, K. Scaman, R. Vidal, L. Kameni, and M. Lorenzi. 2024. SIFU: Sequential informed federated unlearning for efficient and provable client unlearning in federated optimization. In *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*. PMLR, 3457–3465.
- [3] X. Gao, X. Ma, J. Wang, Y. Sun, B. Li, S. Ji, P. Cheng, and J. Chen. 2024. VERiFi: Towards verifiable federated unlearning. *IEEE Trans. Dependable and Secure Computing* (2024).
- [4] Antonio Ginart, Melody Guan, Gregory Valiant, and James Y Zou. 2019. Making AI forget you: Data deletion in machine learning. *Advances in neural information processing systems* 32 (2019).
- [5] T. T. Huynh, T. B. Nguyen, P. L. Nguyen, T. T. Nguyen, M. Weidlich, Q. V. H. Nguyen, and K. Aberer. 2024. Fast-FedUL: A training-free federated unlearning with provable skew resilience. In *Joint Eur. Conf. Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*. Springer, 55–72.
- [6] Yuheng Leng, Lei Xu, Jianghua Liu, Xiaojun Zhang, Lin Mei, Youyang Qu, and Chungen Xu. 2025. FedSSU: flexible and efficient decentralized unlearning for federated learning. *The Journal of Supercomputing* 81, 8 (2025), 986.
- [7] Shenghui Li, Edith Ngai, and Thiemo Voigt. 2021. Byzantine-robust aggregation in federated learning empowered industrial IoT. *IEEE Transactions on Industrial Informatics* 19, 2 (2021), 1165–1175.
- [8] G. Liu, X. Ma, Y. Yang, C. Wang, and J. Liu. 2021. Federaser: Enabling efficient client-level data removal from federated learning models. In *Proc. IEEE/ACM 29th Int. Symp. Quality of Service (IWQoS)*. IEEE, 1–10.
- [9] Ziyao Liu, Yu Jiang, Weifeng Jiang, Jiale Guo, Jun Zhao, and Kwok-Yan Lam. 2024. Guaranteeing data privacy in federated unlearning with dynamic user participation. *IEEE Transactions on Dependable and Secure Computing* (2024).
- [10] Thanh Tam Nguyen, Thanh Trung Huynh, Zhao Ren, Phi Le Nguyen, Alan Wee-Chung Liew, Hongzhi Yin, and Quoc Viet Hung Nguyen. 2025. A survey of machine unlearning. *ACM Transactions on Intelligent Systems and Technology* 16, 5 (2025), 1–46.
- [11] Nicolò Romandini, Alessio Mora, Carlo Mazzocca, Rebecca Montanari, and Paolo Bellavista. 2024. Federated unlearning: A survey on methods, design guidelines, and evaluation metrics. *IEEE Transactions on Neural Networks and Learning Systems* (2024).
- [12] Dian Shi, Liang Li, Rui Chen, Pavana Prakash, Miao Pan, and Yuguang Fang. 2022. Toward energy-efficient federated learning over 5G+ mobile devices. *IEEE Wireless Communications* 29, 5 (2022), 44–51.
- [13] Haofeng Sun, Hui Tian, Wanli Ni, Jingheng Zheng, Dusit Niyato, and Ping Zhang. 2024. Federated low-rank adaptation for large models fine-tuning over wireless networks. *IEEE Transactions on Wireless Communications* (2024).
- [14] Ayush K Tarun, Vikram S Chundawat, Murari Mandal, and Mohan Kankanhalli. 2023. Fast yet effective machine unlearning. *IEEE Transactions on Neural Networks and Learning Systems* 35, 9 (2023), 13046–13055.
- [15] Zhibo Xing, Zijian Zhang, Ziang Zhang, Zhen Li, Meng Li, Jiamou Liu, Zongyang Zhang, Yi Zhao, Qi Sun, Liehuang Zhu, et al. 2025. Zero-Knowledge Proof-Based Verifiable Decentralized Machine Learning in Communication Network: A Comprehensive Survey. *IEEE Communications Surveys & Tutorials* (2025).
- [16] X. Zhong et al. 2025. Unlearning through knowledge overwriting: Reversible federated unlearning via selective sparse adapters. In *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*. IEEE.